(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH] JULY-2023

Paper Code: ETCS-304 ***Subject: Operating Systems***

Time: 3 Hours ***Maximum Marks:75***

Note: Attempt five questions in all including.Qno.1 which is compulsory. Select one question from each unit.

Q1 **Attempt any Five** **[5x5=25]**

(a) What is the meaning of the term *busy waiting*? What other kinds of waiting are there in an operating system? Can busy waiting be avoided altogether? Explain your answer.

(b) What is multiprogramming operating system? How it is different from Multiprocessing. Explain.

(c) Explain Process Control Block with the help of suitable diagram in detail.

(d) Explain race condition with suitable example?

(e) What are the four conditions that must be present for a deadlock to be possible?

(f) What are various file attributes?

UNIT-I

Q2 (a) Explain the layered approach of Operating System? What are main task of OS? **[6]**

(b) Given Memory Partitions of 100K,500K,200K,300K and 600K (in order), how would each of the first fit, best fit, and worst fit algorithms place processes of 212K,417K,112K and 426K (in order)? Which algorithms make the most efficient use of memory? **[6.5]**

Q3 (a) Explain the difference between logical and physical addresses. **[3]**

(b) Why are pages sizes always power of 2? **[3]**

(c) Consider there are 4 frames allocated to a process and the page reference string is:

7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1

Calculate the number of page faults for the FIFO and LRU page replacement algorithms. **[6.5]**

UNIT-II

Q4 (a) Consider the following set of processes with the CPU burst & arrival time in milliseconds. **[6.5]**

Process	Arrival Time	Burst Time
A	0	12
B	1	7
C	2	9
D	3	4
E	4	11

Draw the Gantt chart & find:

P.T.O.